Tobias Mwangi-Hartley

PROFESSIONAL SUMMARY

Data Scientist with experience developing and evaluating forecasting and churn models, customer segmentation, and experimentation tools. Uses Python, SQL and R to build analytical solutions for operational and product decisions. Experience includes scheduled notebook workflows, batch scoring, feature engineering, and automated tests.

EXPERIENCE

Data Scientist | Thistlebox | 2022-Present

- Developed a six-week weekly demand forecast by recipe and region using LightGBM with calendar, price and menu-position features. The scheduled notebook writes forecasts to BigQuery for operations reporting in Looker; WAPE improved from 23% to 14%, with an estimated 9% reduction in food waste.
- Built an XGBoost churn propensity model scored weekly in BigQuery, with retention offers sent to top-decile customers. A 10% holdout validation found churn was 2.1 percentage points lower in the treated group.
- Developed a Streamlit sequential A/B testing calculator for planning and analysing experiments; used by approximately 25 people in product and growth. Wrote the app and its Dockerfile; the data platform team handles deployment to Cloud Run.
- Since 2025, writing pytest unit tests for forecasting feature code; coverage is approximately 60%.
- Delivered the internal training “Stats for PMs” three times.

Junior Data Analyst | Firth Energy Retail | 2020-2022

- Segmented customers and analysed tariffs using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team approximately two days each month.

PROJECT

Image Classification Course Project | Personal | 2025

- Fine-tuned a small CNN on a public food images dataset as part of an online deep learning course; achieved approximately 80% accuracy. Learning project.

EDUCATION

MSc in Statistics

- Dissertation on Bayesian hierarchical models.

SKILLS

Programming and analysis: Python, pandas, scikit-learn, SQL, R, R Markdown
Machine learning and statistics: LightGBM, XGBoost, SHAP, statsmodels, feature engineering, forecasting, churn modelling, experimentation, Bayesian hierarchical models
Data and deployment tools: BigQuery, Vertex AI Workbench, dbt, Looker, Streamlit, Docker
Testing and deep learning: pytest, PyTorch, torchvision (course project)